

Day 20 – CI/CD Pipelines with GitHub Actions & AWS

Introduction

CI/CD automates software building, testing, and deployment.

Continuous Integration

Developers push code and automated tests run automatically.

Continuous Deployment

Applications deploy automatically after successful testing.

GitHub Actions

GitHub Actions automates workflows directly from GitHub.

AWS DevOps Services

CodePipeline, CodeBuild, CodeDeploy, CloudWatch.

Real-World Project

Automated deployment pipeline using GitHub Actions and AWS.

Security

IAM Roles, GitHub Secrets, encrypted credentials.

Monitoring

CloudWatch tracks deployments and application health.

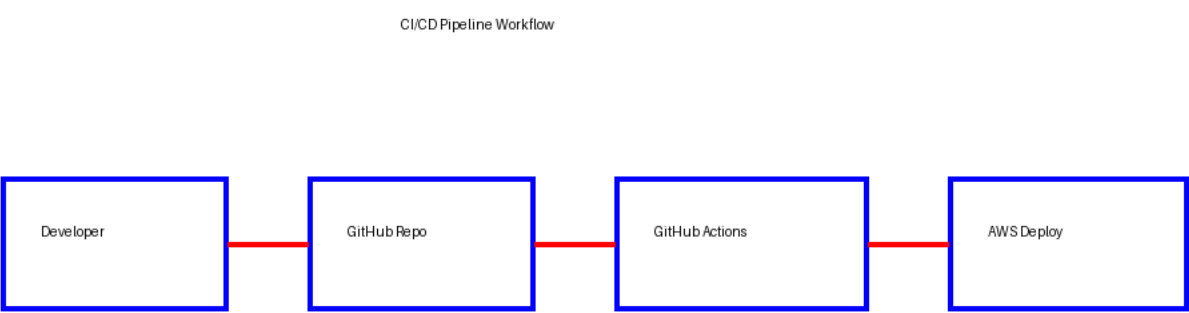
Advantages

Faster deployments and DevOps automation.

Use Cases

Kubernetes deployments, SaaS delivery, enterprise automation.

CI/CD Workflow Diagram



AWS DevOps Services

Service	Purpose
GitHub Actions	Workflow automation
CodePipeline	CI/CD orchestration
CodeDeploy	Deployment automation
CloudWatch	Monitoring & logging

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